GATE

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EC : ELECTRONICS AND COMMUNICATIONS

Networks

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Topic 1 : Network Elements

NETWORK ELEMENTS

Current : The phenomenon of transferring charge from one point in a circuit to another is described by the term electric current. An electric current may be defined as the time rate of net motion of electric charge across a cross sectional boundary.

$$i = \frac{dq}{dt} \text{ Amp}$$

If the charge *q* is given in coulombs and the time *t* is measured in seconds, the current *i* is measured in amperes.

Voltage : The voltage is defined as work per unit charge.

$$v = \frac{w}{q} \text{ Volts}$$

If w is the work (or energy) in joules and q is the charge in coulombs, the voltage *v* is in volts.

***Note* :** If the potential is multiplied by the current, the product is called power *p*.

$$p = v\,i \text{ Watts}$$

Energy (w) as a function of power is found by integrating above equation. Thus total energy at any time t is the integral.

$$w = \int_{-\infty}^{t} p\,dt$$

THE RESISTANCE

Physical phenomenon : While passing through a material the amount of energy lost by the electrons due to collisions with atomic particles, is related to the physical properties of a particular substance.

Field interpretation: The change in energy per unit of charge causes a force per unit charge or electric field. This effect may be interpreted in terms of a field in the direction of current through the conducting substance.

$$J = \sigma E \qquad \text{....(1)}$$

where, J = current density in A/m^2;
E = field intensity along the conducting substance V/m;
σ = conductivity.

Circuit Interpretation: If the cross section of the conductor is uniform, the current and current density are related by the equation

$$I = \int_s J \cos\theta \, ds = JS \qquad \text{....(2)}$$

where S = cross sectional area.

The electric field is uniform and directed along the length of the wire;

$$v = E\ell \qquad \text{....(3)}$$

$$v = \int_{\ell_1}^{\ell_2} E\cos\theta d\ell$$

Substituting eqn.(2) & (3) into (1)

$$v = \left(\frac{\ell}{\sigma S}\right) i$$

The quantity $\left(\frac{\ell}{\sigma S}\right)$ is given then name '**resistance**' (R).

$$\therefore \quad \boxed{v = Ri}$$

This is also written as

$$v = \frac{1}{G} i$$

or $\boxed{i = Gv}$

where G is known as "**conductance**".

The unit of "resistance" (R) is ohm (Ω) and the unit of "Conductance" (G) is mho (℧) or siemens.

The resistance of a material depends upon nature of the material, geometry of the material and also on temperature.

$$R = \rho\frac{\ell}{A}\ \Omega$$

where, ρ = resistivity, in Ω–m or Ω–cm

ℓ = length of material in m or cm

A = cross–sectional area in m^2 or cm^2

$$\boxed{R_t = R_0\,(1+\alpha\Delta t) \quad \Omega}$$

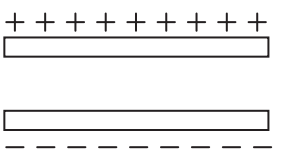
where, R_t = resistance at t°C

R_0 = resistance at reference temperature (0°C)

α = temperature coefficient of resistance

THE CAPACITANCE

Physical phenomenon: The spatially spaced substances having charge between them experiences force. This force is of such nature that "like charges repels" and "unlike charges attract"



$$\boxed{F = \frac{q_1 q_2}{4\pi\varepsilon r^2}}$$

where F = Force in Newtons

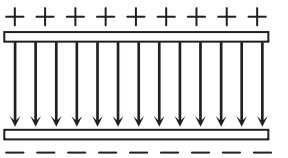
r = separation between charges in meters.

ε = permittivity, 8.854×10^{-12} F/m

q_1, q_2 = charges measured in Coulombs.

Note **:** Above equation is valid only for point charges.

Field Interpretation: The force per unit charge (a vector quantity since force is a vector quantity) is called an electric field of value $E = \frac{F}{q}$.



For the parallel plates voltage v = Ed,
where d is the separation of the plates.
Thus we have

$$v = \left(\frac{d}{\varepsilon A}\right) q$$

where, $E = \frac{q}{\varepsilon A}$, A = plate area and q = charge on plate.

$$v = Dq$$

where, D is defined as '**elastance**'. The reciprocal of D is defined as '**capacitance**' C.

$$\therefore \quad q = Cv$$

So, for the parallel plates

$$\boxed{C = \frac{\varepsilon A}{d}} \text{ Farad (F)}$$

We know that

$$i = \frac{dq}{dt} \text{ Amp} = C\frac{dv}{dt} \text{ Amp.}$$

> Capacitance C does not vary with time.
>
> $$i = C\frac{dv}{dt} \text{ Amp}$$

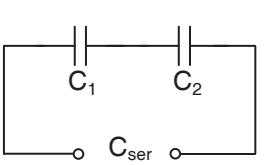
If however, C is not constant but varies as a function of time, the current i is

$$i = \frac{d}{dt}(Cv) = C\frac{dv}{dt} + v\frac{dC}{dt}$$

Similarly,

$$v = \frac{q}{C} = \frac{1}{C}\int_{-\infty}^{t} i\, dt$$

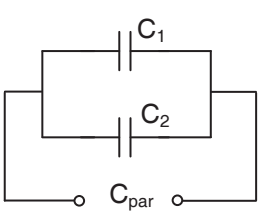
Capacitors in series :



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{C_1 + C_2}{C_1 C_2}$$

$$\therefore \quad C_{ser} = \frac{C_1 C_2}{C_1 + C_2}$$

Capacitors in parallel :



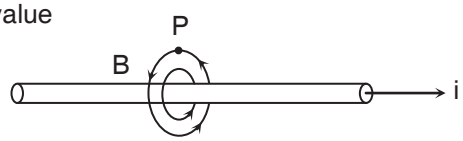
$$C_{par} = C_1 + C_2$$

THE INDUCTANCE

Physical phenomenon: The force between two charged substances depends on the time rate of flow of charge (current).

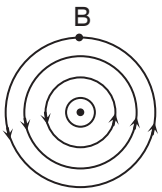
Field interpretation : The above phenomenon can be interpreted in terms of force per unit magnetic pole at all points in space.

The force is directed at right angles to the current carrying conductor. The magnetic field density B, the force per unit magnetic pole, of value



$$dB = \frac{\mu i \cos\alpha \,.\, d\ell}{4\pi r^2}.$$

where, μ = magnetic permeability
i = current in amperes



Magnetic flux ϕ is defined as

$$\phi = \int_s B \cos\theta \, d\theta$$

ϕ = magnetic flux in Weber
θ = angle between the surface of integration and field density B.

If the currents in each of N conductors, represented are in such a direction that the fluxes add, then $N\phi$ flux linkages are said to exist. If, however, ϕ lines of flux link N_1 conductors, ϕ_2 line link N_2 conductors and so forth, the total flux linkages is

$$\psi = \sum_{j=1}^{n} N_j \phi_j$$

$$\psi = N \int_s B \cos\theta \, ds$$

$$\psi = \left[N \int \left(\int \frac{\mu \cos\alpha \, d\ell}{4\pi r^2} \right) ds \right] i$$

$$= L\, i$$

where L is an 'inductance' measured in Henry (H).

As per Faraday's law, the voltage is induced in a coil if there is rate of change of flux.

$$v = -N\frac{d\phi}{dt} = \frac{d}{dt}(Li)$$

$$= L\frac{di}{dt}$$

and $$i(t) = \frac{1}{L}\int_{-\infty}^{t} v \, dt$$

or $$i(t) = \frac{1}{L}\int_{0}^{t} v dt + i(0)$$

The Henry (H) is the MKS unit for inductance power in inductor

$$P = vi$$

$$= L i \frac{di}{dt}$$

Energy stored in an inductor (w)

$$w = \int p \, dt$$

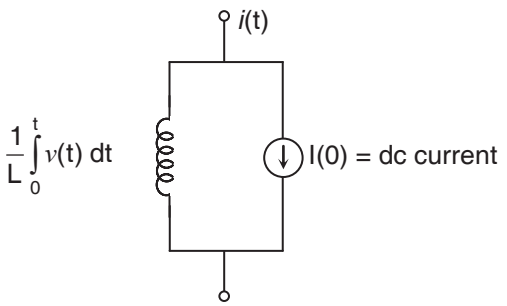
$$= \int L i \frac{di}{dt} dt = \int L i \, di$$

$$= \frac{L i^2}{2} \text{ joules}$$

From the equation (4), the current in an inductor is the sum of two currents

$$i(t) = \frac{1}{L}\int_0^t v(t)\, dt + i(0) \qquad \ldots(4)$$

Hence the total current in an inductor can be considered as two currents in parallel.



For *passive element* energy absorbed $[\, w(t) \,] = \int_{-\infty}^{t} v(t)\, i(t)\, dt \geq 0$

For *active element* energy delivered $[\, w(t) \,] = \int_{-\infty}^{t} v(t)\, i(t)\, dt \leq 0$

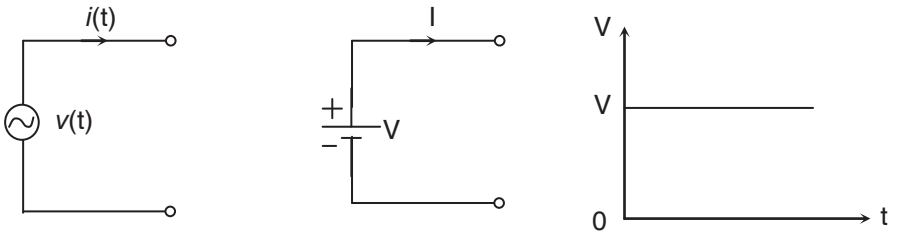
- Passive elements are those that absorb power or energy. Active elements deliver power or energy.
- Negative sign indicates an active element delivering power and a positive sign indicates a passive element absorbing power.

Inductors and capacitors absorb power for t > 0 and return to the source when switched off. The energy equations for them are $\frac{1}{2}Li^2$ & $\frac{1}{2}Cv^2$ respectively. So that for '+' or '–' voltage and current, energy is always positive.

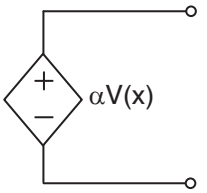
ENERGY SOURCES

There are two types of Energy Sources:
(1) Ideal voltage source (2) Ideal current source

1. Ideal Voltage Source



Independent Ideal Voltage Source

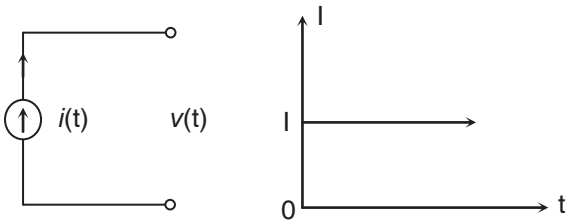


Dependent Ideal Voltage Source

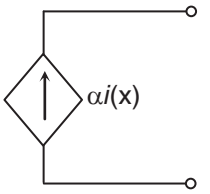
- An ideal voltage maintains a terminal voltage $v(t)$ regardless of the value of the current through its terminals.
- In practical voltage source, the voltage across the terminals of the source keep falling as the current through it increases.

2. Ideal Current Source

An ideal current source is a two–terminal element which maintains a current i(t) flowing through its terminals, regardless of the value of the terminal voltage.



Independent Current Source



Dependent Ideal Current Source

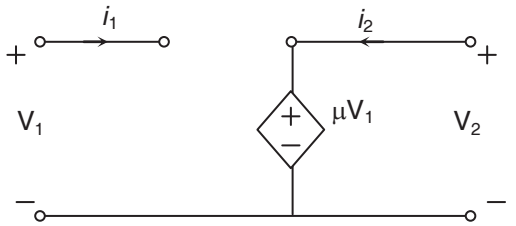
The associated current voltage sign convention is such that the current flows out of the terminal marked '+' i.e. along the direction of arrow.

In practical current source, the current through the source decreases as the voltage across it increases.

CONTROLLED SOURCES

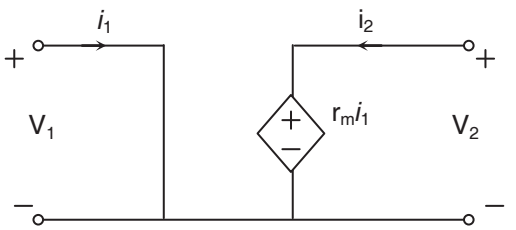
Dependent or controlled sources are of the following type :

1. *Voltage Controlled Voltage Source* (VCVS) :



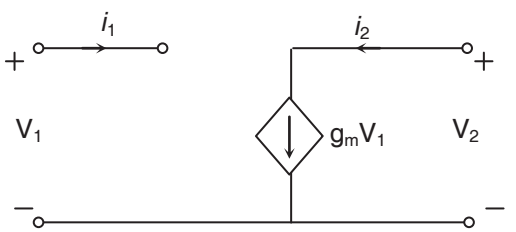
(VCVS)

2. *Current Controlled Voltage Source* (CCVS) :



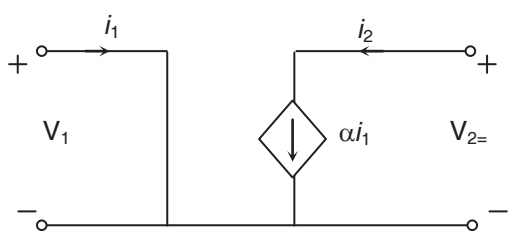
(CCVS)

3. *Voltage Controlled Current Source* (VCCS) :



(VCCS)

4. *Current Controlled Current Source* (CCCS) :



(CCCS)

LIST OF FORMULAE

- *Current*

$$i = \frac{dq}{dt} \text{ Amp}$$

If the charge q is given in coulombs and the time t is measured in seconds, the current is measured in amperes.

- *Voltage*

$$v = \frac{w}{q} \text{ Volts}$$

If w is the work (or energy) in joules and q is the charge in coulombs, the voltage v is in volts.

- *Power*

$$p = vi \text{ Watts}$$

- *Current density*

$$J = \sigma E$$

where J = current density in A/m^2;
E = field along the conducting substance V/m;
σ = conductivity.

- *Force*

$$F = \frac{q_1 q_2}{4\pi\varepsilon r^2}$$

where F = Force in Newton
r = separation between charges in meters.
ε = permittivity, 8.854×10^{-12} F/m
q_1, q_2 are the charges measured in coulombs.

- *Capacitor*

For the parallel plates voltage

$$v = Ed,$$

where d is the separation of the plates.

Thus we have, $v = \left(\frac{dq}{\varepsilon A}\right)$

where $E = \frac{q}{\varepsilon A}$
A = plate area
q = charge on plate.
$v = Dq$
where D is defined as 'elastance'.
The reciprocal of D is defined as 'capacitance' C.
$\therefore \quad q = Cv$

So, for the parallel plates, $C = \frac{\varepsilon A}{d}$ Farad (F)

Energy stored in capacitor = $\frac{1}{2}Cv^2$

Capacitors in Series :

$$C_{ser} = \frac{C_1C_2}{C_1 + C_2}$$

Capacitors in parallel :

$$C_{par} = C_1 + C_2$$

- *Inductor*

  The voltage induced in a coil

$$V = -N\frac{d\phi}{dt} = \frac{d}{dt}(Li) = L\frac{di}{dt} \quad \text{Volts}$$

- *Energy stored in a inductor*

$$w = \int p\, dt = \int Li\frac{di}{dt}dt = \int Li\, di = \frac{Li^2}{2} \text{ joules}$$

- *Energy*

  For Passive Element, $w(t) = \int_{-\infty}^{t} v(t)\, i(t)\, dt \geq 0$

  For Active Element, $w(t) = \int_{-\infty}^{t} v(t)\, i(t)\, dt \leq 0$

LMR (LAST MINUTE REVISION)

- An electric current may be defined as the time rate of net motion of electric charge across sectional boundary.
- The voltage is defined as work per unit charge.
- Total energy at any time t is the integral $w = \int_{-\infty}^{t} p\, dt$
- Power is also defined as, it is a certain amount of energy used over a certain length of time (rate at which work is done).
- The quantity $\left(\frac{\ell}{\sigma S}\right)$ is given then name 'resistance' (R).
- $i = Gv$ where G is known as "conductance".
- The unit of "resistance" (R) is ohm (Ω) and the unit of "Conductance" (G) is mho ($\mho$) or siemens.
- The resistance of a material depends upon nature of the material, geometry of the material and also on temperature.
- Passive elements are those that absorb power or energy. Active elements deliver power or energy. So negative sign indicates an active element delivering power and a positive sign indicates a passive element absorbing power.
- When current enters positive terminal and leaves negative terminal, power is absorbed and in other situation power is delivered.

❑ ❑ ❑ ❑ ❑ ❑

ASSIGNMENT – 1

Duration : 45 mins **Marks : 30**

Q1 to Q6 carry one mark each

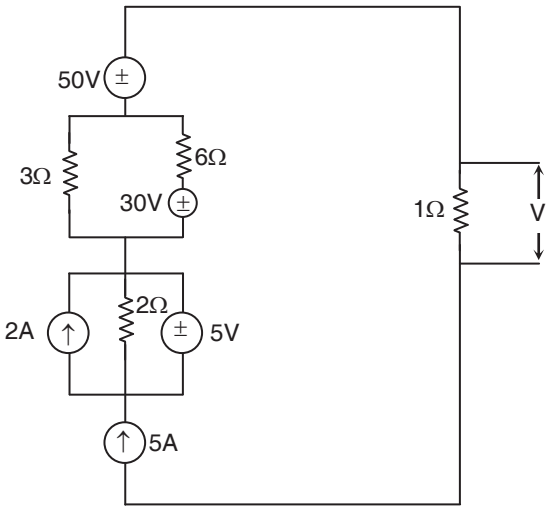
1. A metal wire has a uniform cross–section 'A' length 'ℓ' and resistance 'R' between its 2 end points. It is uniformly stretched so that its length becomes 'αL'. The new resistance is

(A) α R (B) α^2 R
(C) $\sqrt{\alpha}$ R (D) e^{α} R

2. A 12 V automobile battery is connected to a 5 A headlight. The energy consumed in 1 hr of operation is _____.

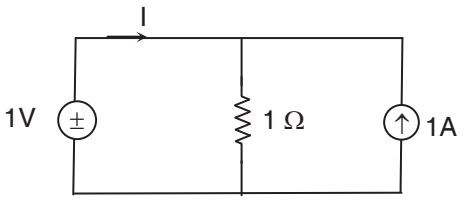
(A) 36 kJ (B) 216 kJ
(C) 196 kJ (D) 225 kJ

3. Find the voltage across 1Ω resistor



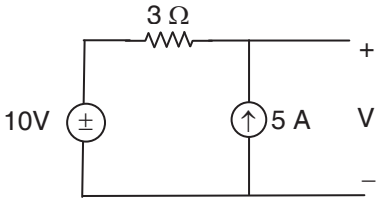
(A) 55 V (B) 90 V
(C) 85 V (D) 5 V

4. The current I supplied by dc voltage source is



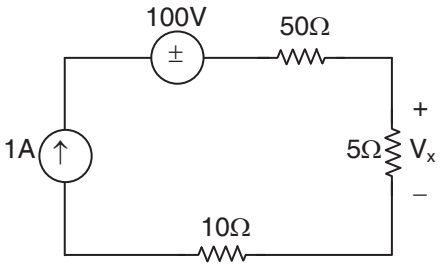
(A) 0 A (B) –1 A
(C) 1 A (D) 2 A

5. Find the voltage 'V'.



(A) 5 V (B) 0 V
(C) 25 V (D) –5 V

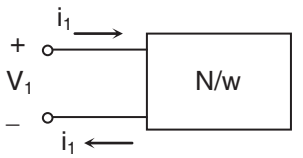
6. Find the V_x.



(A) 37.5 V (B) 5 V
(C) 32.5 V (D) 100 V

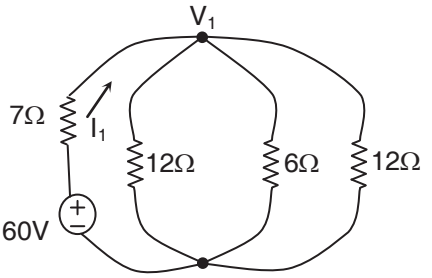
Q7 to Q18 carry two marks each

7. If resistor R is function of current i in the figure shown i.e. $R = i_1/2$. Find the value of i_1 in terms of V_1.



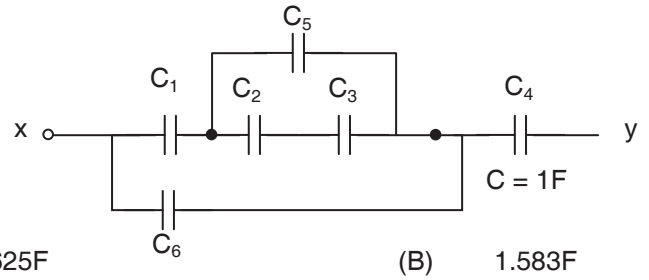
(A) $2\ V_1$ (B) $V_1/2$
(C) $\sqrt{2V_1}$ (D) $2\sqrt{V_1/2}$

8. The no. of equations to solve for V_1 and value of V_1 are respectively,



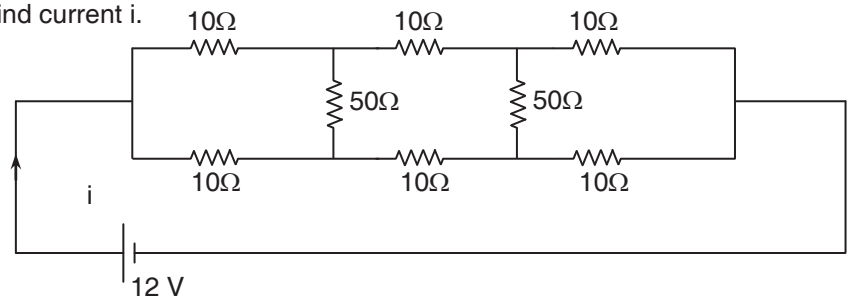
(A) 2, 18 V (B) 2, 6 V
(C) 1, 18 V (D) 1, 6 V

9. The equivalent capacitance across x–y is,



(A) 1.625F (B) 1.583F
(C) 0.583F (D) 0.615F

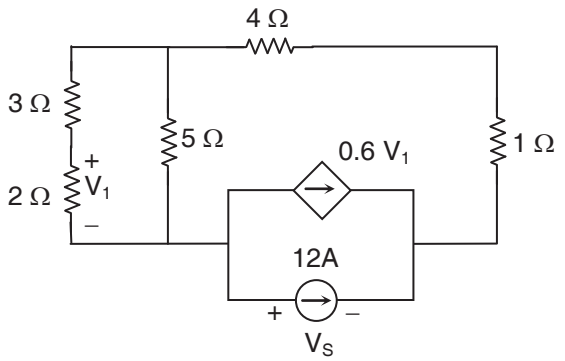
10. Find current i.



(A) 4/5 A (B) 5/4 A
(C) 1.23 A (D) 1.32 A

11. For a given voltage, four heating coils will produce maximum heat when connected
(A) All in parallel
(B) All in series
(C) With two parallel pairs in series
(D) One pair in parallel with the other two in series

12. In the given circuit, voltage vs across the current source is

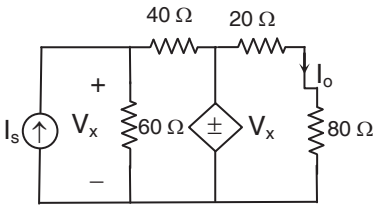


(A) 25 V (B) 225 V
(C) –225 V (D) –22.5 V

13. A 10 V battery with 1 Ω internal resistance is connected across a non linear load whose V–I characteristic is given as $7\,I = V^2 + 2V$. The current delivered by the battery is _____

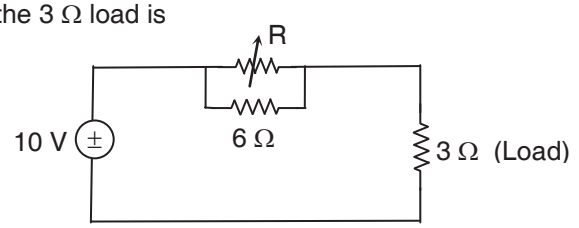
(A) 5 A (B) 14 A
(C) 35 A (D) 20 A

14. In the given circuit, the current gain $\left|\frac{I_o}{I_s}\right|$ is equal to



(A) –0.15 (B) 0.15
(C) 0.30 (D) –0.30

15. In the circuit below, the value of R required for the maximum transfer of power to the 3 Ω load is



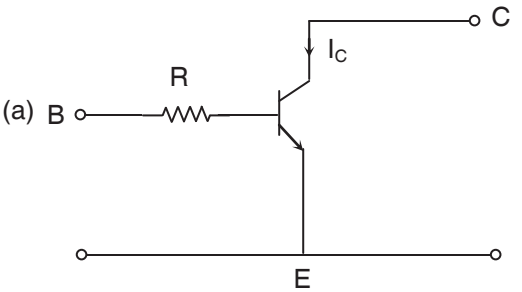
(A) Zero (B) 3 Ω
(C) 6 Ω (D) ∞

16. The list I below lists the various dependent sources whereas list II gives real life examples of these sources. Match the following using the codes given below.

List I **List II**

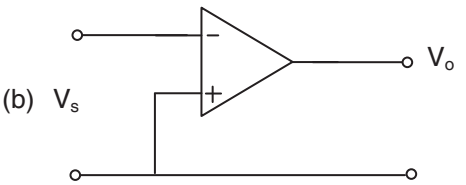
1. VCVS

(a)

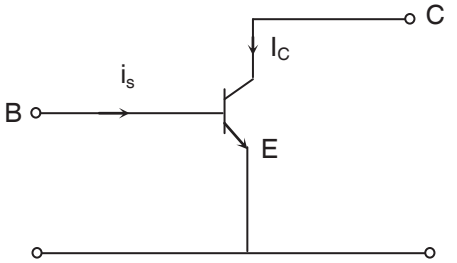


2. CCVS

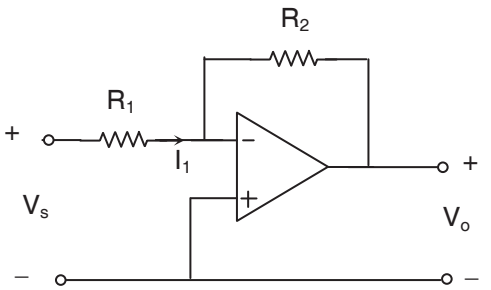
(b)



3. VCCS (c)



4. CCCS (d)

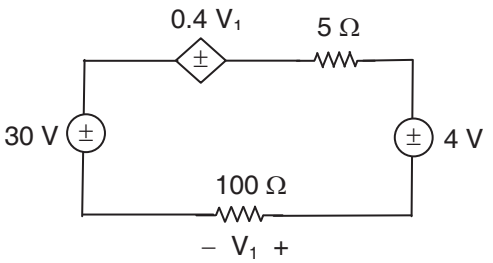


Codes:

(A) 1 – c, 2 – a, 3 – b, 4 – d (B) 1 – b, 2 – d, 3 – a, 4 – c

(C) 1 – a, 2 – b, 3 – d, 4 – c (D) 1 – b, 2 – a, 3 – c, 4 – d

17. How much power is absorbed by dependent source in the following circuit ?



(A) 3.2W (B) –6.4 W

(C) –3.2 W (D) 6.4 W

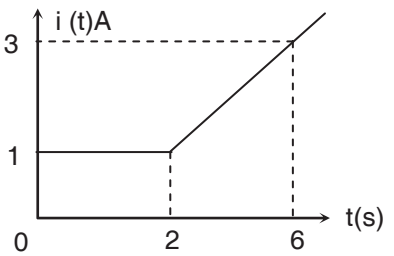
18. The current entering a circuit element is shown in the figure. The charge that passes through the element from t = 0 to t = 6 will be

(A) 4 C

(B) 6 C

(C) 12 C

(D) 9 C



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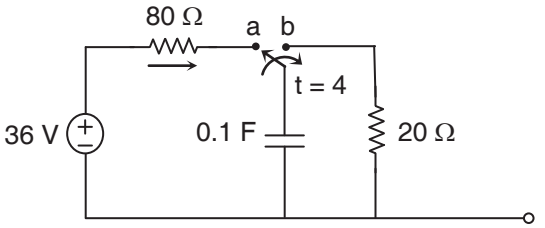
TEST PAPER – 1

Duration : 30 mins **Marks : 25**

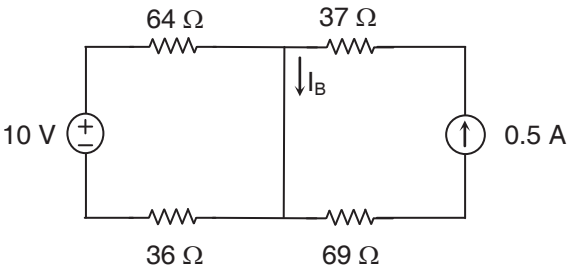
Q1 to Q5 carry one mark each

1. Consider the circuit, the switch moves from 'a' to 'b' at t = 4s. Find the voltage across the capacitor at t = 10 s.



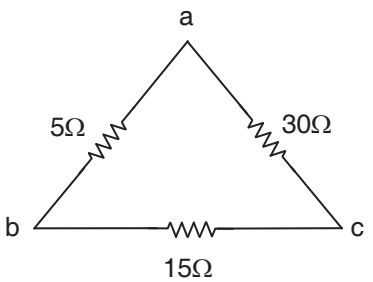
(A) 1.8 V (B) –3.6 V
(C) –1.8 V (D) 3.6 V

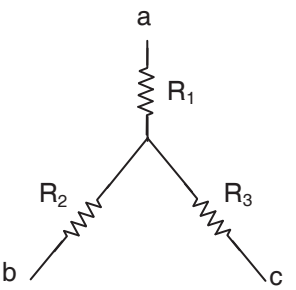
2. The current I_B in the circuit is



(A) 0.6 A (B) –0.6 A
(C) 0.4 A (D) –0.8 A

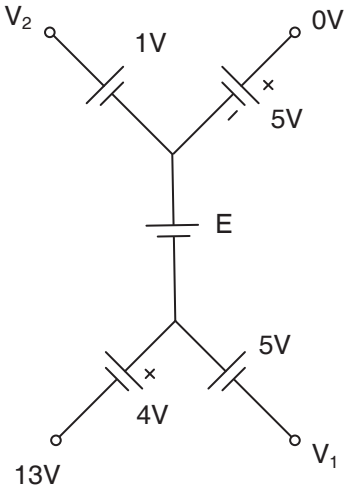
3. Figure (a) is redrawn as (b). Find R_1, R_2, R_3.





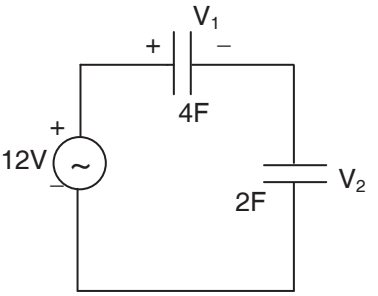
(A) 1.5, 3, 9 (B) 9, 3, 1.5
(C) 3, 9, 1.5 (D) 3, 1.5, 9

4. In the circuit, the value of the voltage source E is,



(A) –22V (B) –12V
(C) +12V (D) –10V

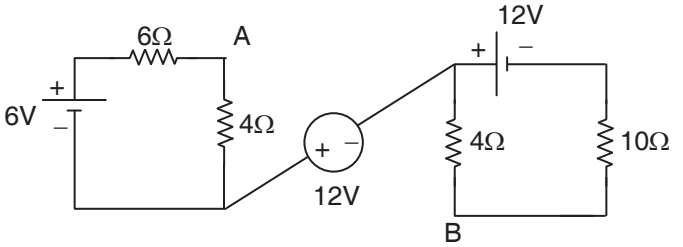
5. The voltage V_1 and V_2 in the given circuit are



(A) 4 V, 8 V
(B) 8 V, 6 V
(C) 6 V, 6V
(D) 8 V, 4 V

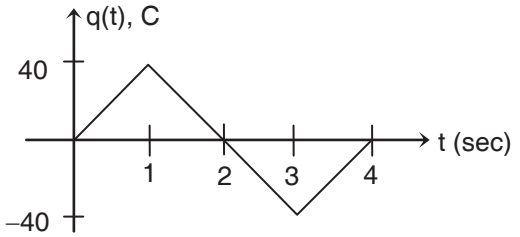
Q6 to Q13 carry two marks each

6. What is the voltage across A and B in the circuit shown in figure?

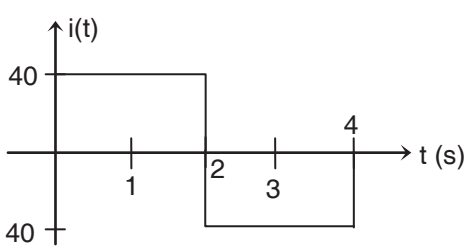


(A) 13.04 V (B) 17.84 V
(C) 12 V (D) 10.96 V

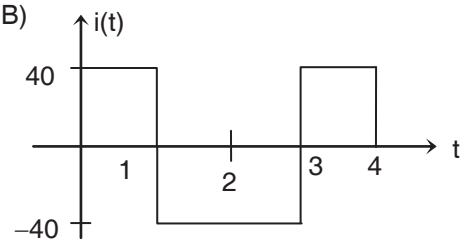
7. The charge passing through a circuit element is sketched as



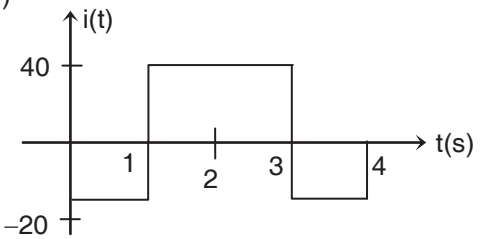
(A)



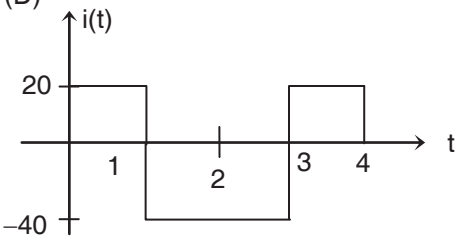
(B)



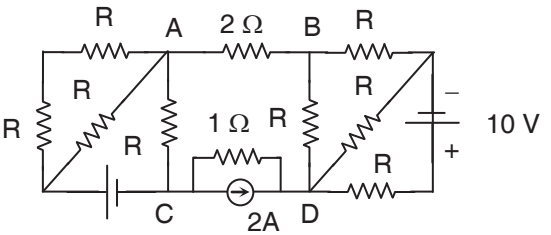
(C)



(D)



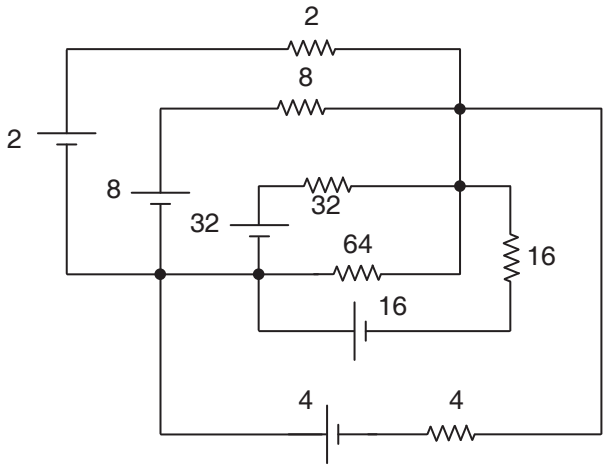
8. If $V_A - V_B = 6$ V, $V_C - V_D = ?$



[Given R = 5 Ω]

(A) –5 V
(B) 2 V
(C) 5 V
(D) 6 V

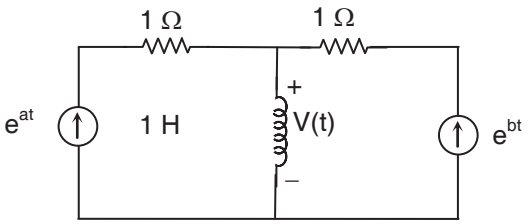
9. Find the voltage across 64 Ω resistor.



(A) $\frac{-64}{63}$ V (B) $\frac{63}{64}$ V

(C) $\frac{64}{63}$ V (D) $\frac{-63}{64}$ V

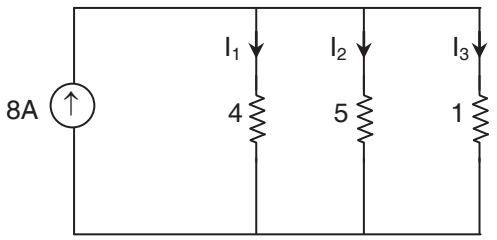
10. In the circuit below, the voltage V(t) is



(A) $e^{at} - e^{bt}$ (B) $e^{at} + e^{bt}$

(C) $ae^{at} + be^{bt}$ (D) $ae^{at} - be^{-bt}$

11. Find currents I_1, I_2 & I_3.



(A) 1.38A, 1.1A, 5.52A (B) 1.38A, 5.5A, 1.1A

(C) 1.38A, 0.9A, 6.38A (D) 1.1A, 5.52A, 1.38A

12. When a dc voltage is applied to an inductor, the current through it is found to build up in accordance with $i = 20\left(1 - e^{-50t}\right)$. After the lapse of 0.02s, the voltage is equal to 2 V. Find the value of inductance.

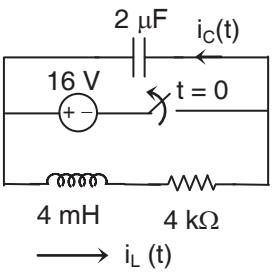
(A) 2 mH (B) 5.43 mH
(C) 1.54 mH (D) 0.74 mH

13. Which of the following statements are true?
i) Power is the rate of change of energy
ii) Energy is the rate of change of power
iii) Energy is nothing but stored work

(A) i only (B) i and iii
(C) ii and iii (D) ii only

Q14(a) & (b) carry two marks each

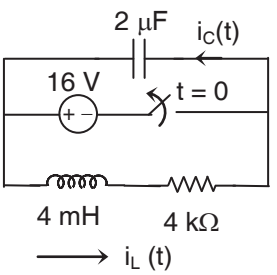
14(a). Consider the circuit.



The value of $i_L(0^+)$ is _____

(A) –4 mA (B) 8 mA
(C) –8 mA (D) 4 mA

14(b). Consider the circuit.



The value of $\frac{di_L(0^+)}{dt}$ is _____

(A) 0 mA/S (B) 4 mA/S
(C) 6 mA/S (D) –4 mA/S

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